

California State University Long Beach

Team Auger

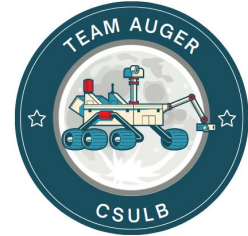


Statement of Work Briefing

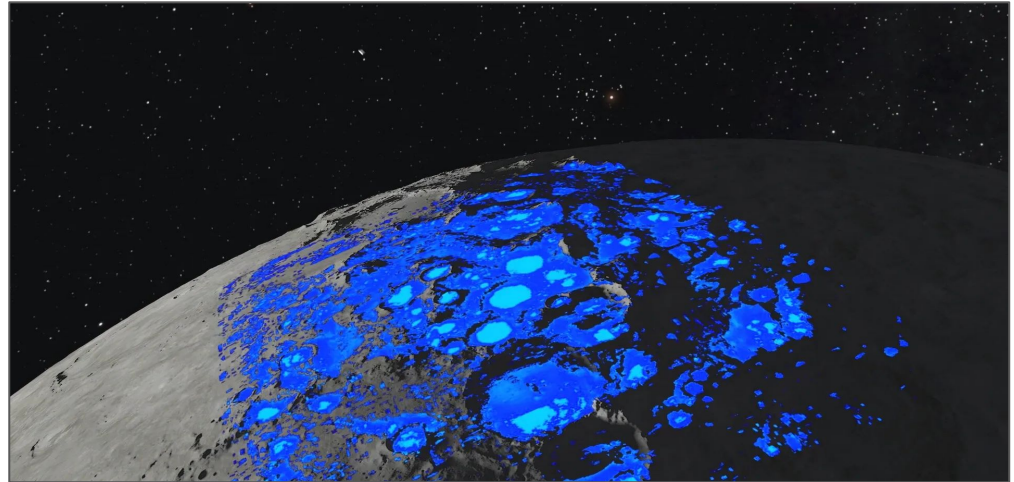
March 8, 2024



Agenda



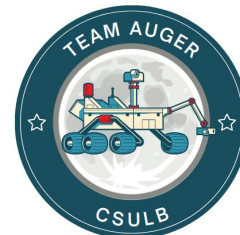
1. Team Introductions
2. Project Overview
3. Changes Since PDR
4. Current Status
5. Scope of Work
6. Task List and Schedule
7. Planned Softwares
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13. Team Leader's Assessment



<https://www.technologyreview.com/2020/10/26/1011228/water-moon-more-accessible-nasa-sofia-lro-glass-micro-cold-trap/>



Team Introduction



Alex Contreras

Team Lead &
Heat Transfer
Lead



Experience in
Power Budgets,
Subsystem
Integration,
Design Reviews

Zachary Tep
Avionics Lead



Experience in
structure,
manufacturing
and propulsion.

**Alondra
Barba**
Structures Lead



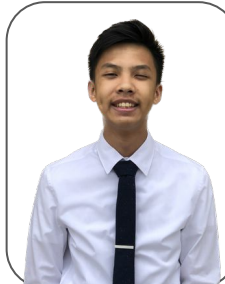
Experience in
structure,
design, and
manufacturing

David Guerra
GNC Lead



Experience in
structures, fluids
mechanics and
aerodynamics.

Erick Nguyen
Mechanical
Design lead



Experience in
mechanical
design and
structural
analysis

Bryan Ramos
Telecommuni-
cations lead



Experienced in
structural analysis,
fluids and
Stability/control

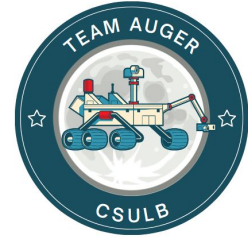
**Damon
Nguyen**
Power Lead



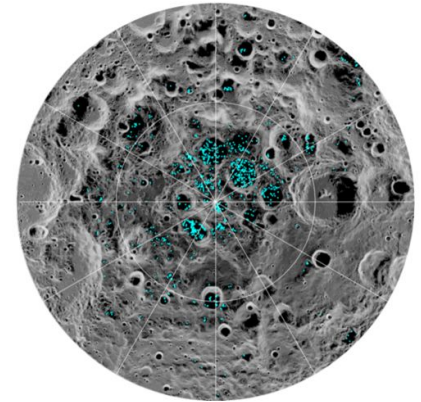
Experience in
thermodynamic,
structures,
aerodynamics,
and fluids.



Project Overview



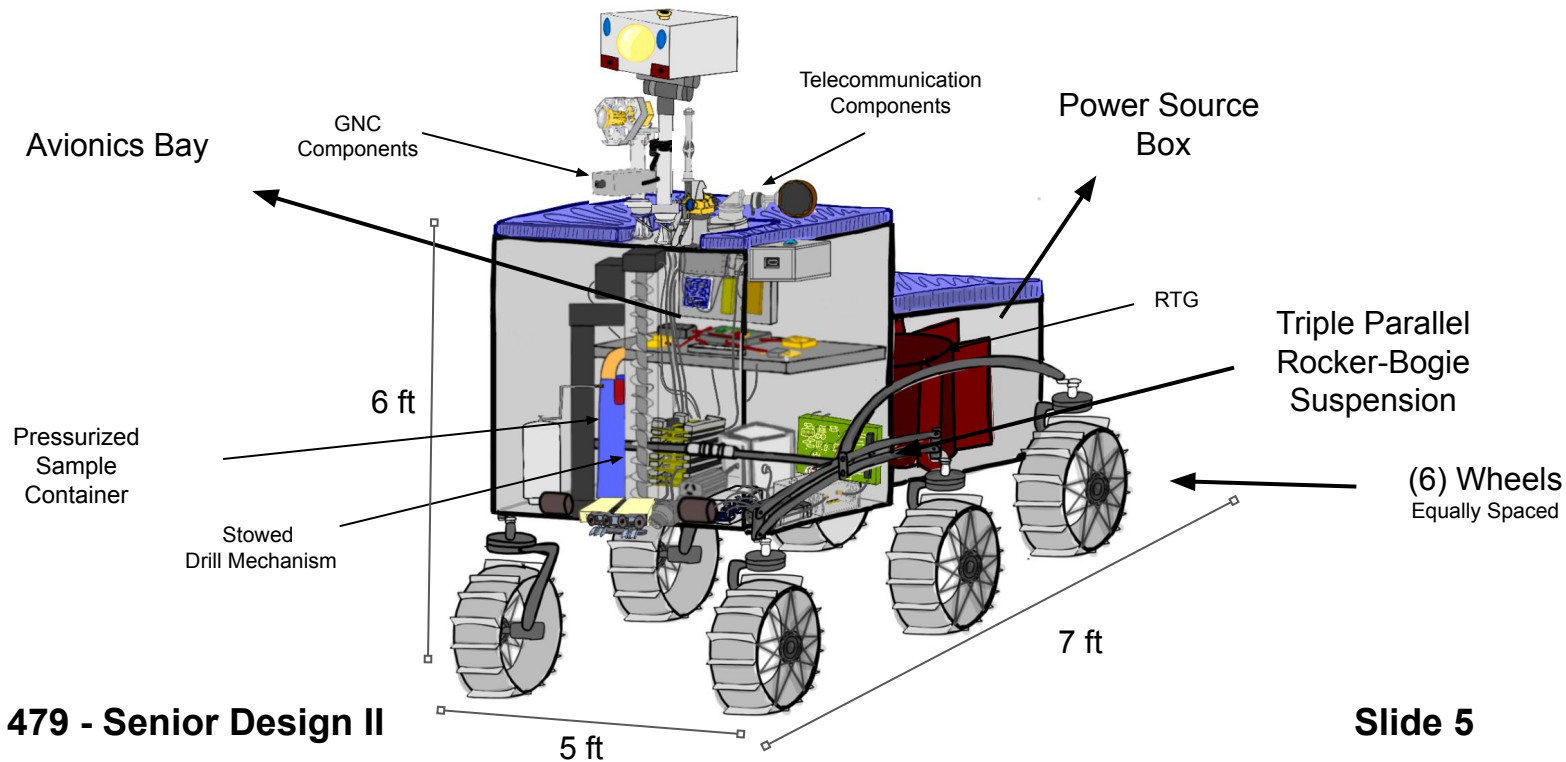
To design an autonomous lunar rover to demonstrate its ability to extract water ice on the lunar surface. The autonomous rover will be engineered to identify areas in the permanently shadowed region with potential water ice deposits. It will then be able to collect and store water ice samples in a pressurized storage tank to avoid sublimation. The rover will be able to withstand the moon's extreme cold temperatures. The rover will aid in the goal of establishing a habitable environment on the lunar surface by demonstrating its ability to extract and store water ice.



The distribution of surface ice (marked in blue) at the Moon's south poles. For more information [visit](#).

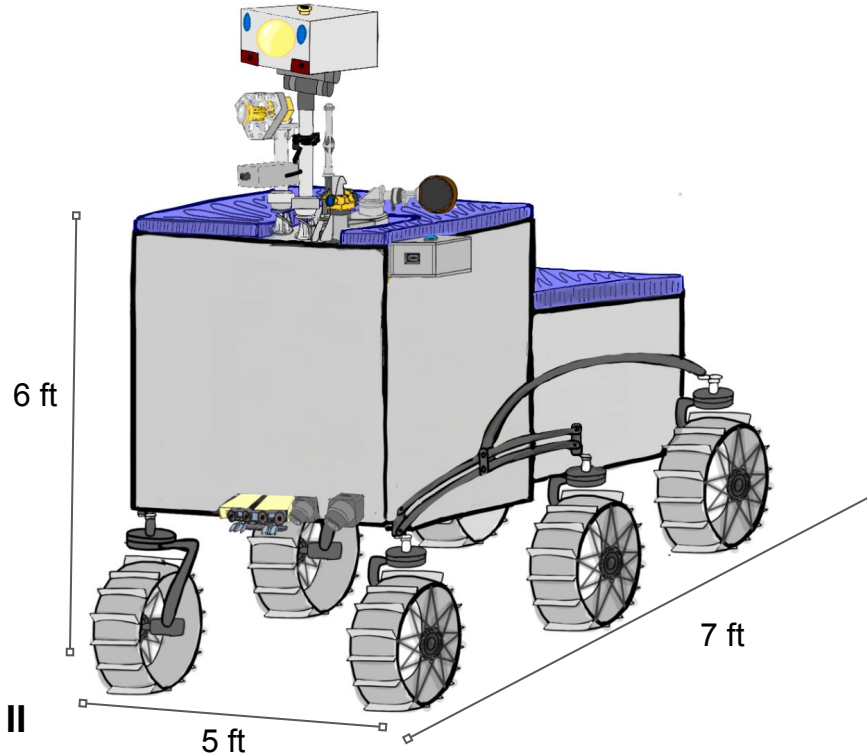
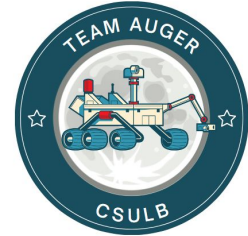


Project Overview



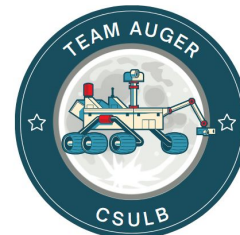


Project Overview





Changes Since PDR

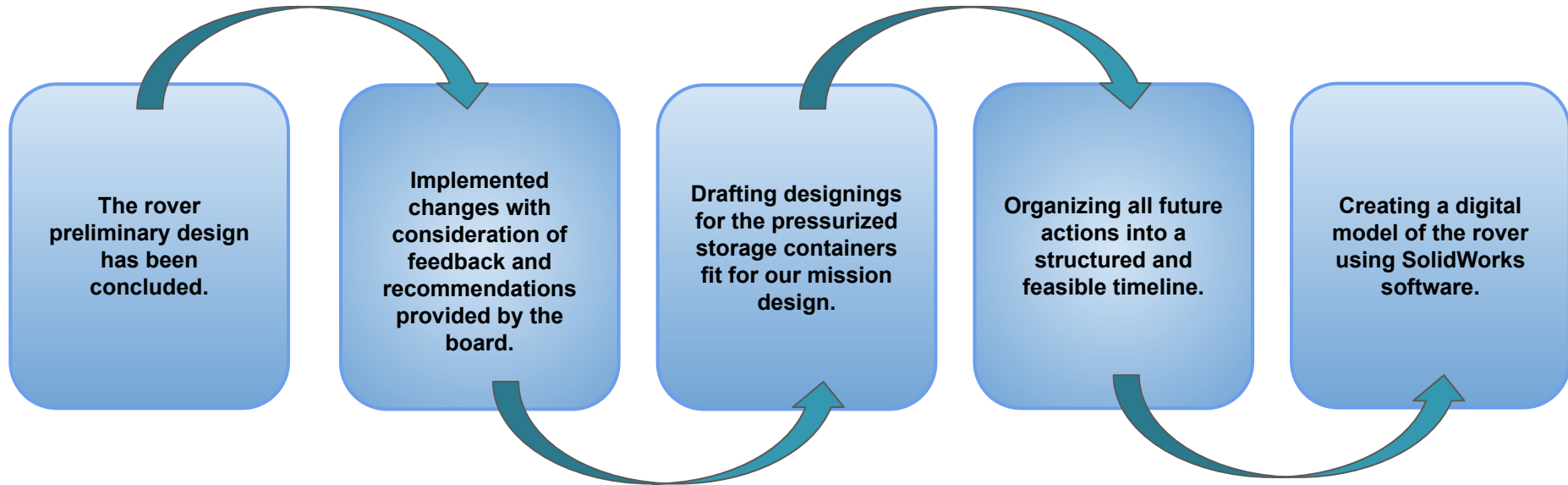
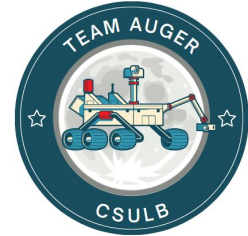


Change #	Relevant Subsystem	Description	Reason for Change	Impact of Change
1	Telecomm	The baseline for our method of communication was using a lander to communicate from the south pole of the moon to Earth, and what our baseline is now, is using a moon base for communication.	Board Recommendation	4
2	Structures	Modifications were made to the wheel design. The wheel was a solid sheet of titanium aluminum with cleats for traction and is now a nickel titanium wire mesh tire with smaller cleats. Changes were made to avoid accumulation of lunar dust on the wheels and optimizing wheel traction and mobility on lunar regolith.	Board Recommendation	3
3	GNC	There was only one autonomous navigation algorithm. There is now a redundancy in order to account for failures or difficulties that the GNC subsystem may encounter	Board Recommendation /industry practice	3

Scale 1-5 : (1=Negligible) (5=Major)

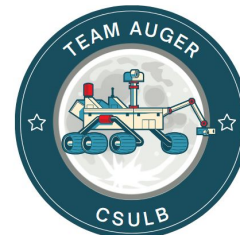


Current Status





Scope of Work



Mission ESD: January 2025

Mission ECD: November 2030

Work Package #1

**Flat-Sat Unit
Assembly**

&
**Flat-Sat Unit
Testing**

Work Package #2

**Engineering
Model
Assembly**

&
**Engineering
Model Testing**

Work Package #3

**Qual & Flight
Unit Parts
Procurement**

Work Package #4

**Comprehensive
Performance
Testing
(CPT)**

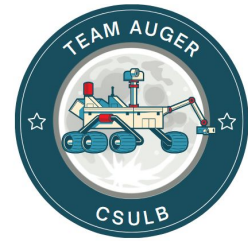
Work Package #5

**Limited Flight
Unit
Performance
Test**



Work Package #1

Flat-Sat Unit Build & Test



Deliverable	Completely built and tested flat-sat that is fully functional and capable of proper communication between each subsystem throughout the entire vehicle.	ESD ECD	Jan 2025 Oct 2025
Description: The flat sat will include and test all of the vehicle's software and verify its operation is fully functional throughout. Analysis of the collected data will then be used to determine if the design needs to be altered.			
Task #	Task Description	Task ESD	Task ECD
1.1	Procure parts for Flat-Sat	Jan 2025	Apr 2025
1.2	Develop and utilize software and hardware for Flat-Sat testing	Apr 2025	May 2025
1.3	Perform integration and interface testing of the Telecomm, Avionics, and GNC systems.	May 2025	Jun 2025
1.4	Collect and analyze data provided from integration and Interface testing	Jun 2025	Aug 2025
1.5	Update design to reflect any necessary adjustments from test results	Aug 2025	Oct 2025



Work Package #2

Engineering Model (EM)

Build & Test

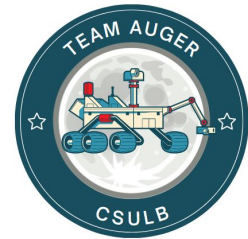


Deliverable	Constructed engineering model, built with commercial parts, and has been fully tested, with collected data that has been analyzed to govern design changes if necessary.	ESD ECD	Nov 2025 Feb 2027
Description: The engineering model will be constructed using commercial parts along with the flat sat. It will undergo rigorous testing in order to verify proper operation and will determine if design changes are needed.			
Task #	Task Description	Task ESD	Task ECD
2.1	Acquire necessary commercial off-the shelf materials and components for our EM	Nov 2025	Jan 2026
2.2	Complete assembly of subsystems for the EM	Jan 2026	Mar 2026
2.3	Perform sub-assembly tests for different subsystems: (Mechanism, Structure, Power)	Mar 2026	May 2026
2.4	Complete assembly of EM	May 2026	Aug 2026
2.5	Perform EM tests under various operating conditions: Environmental, Structural, Performance in a lunar testbed such as The Moonyard	Aug 2026	Oct 2026
2.6	Analyze and publish test results	Oct 2026	Dec 2026
2.7	Re-baselining of EM based on test results as needed	Jan 2027	Feb 2027



Work Package #3

Qual & Flight Parts Procurement



Deliverable	Necessary hardware from manufacturers procured and has been prepared for the assembly of the qualification and flight unit.	ESD ECD	Feb 2027 Nov 2027
Description: Obtain all necessary hardware for the qualification unit and flight unit from manufacturers to have ready and prepared for assembly in a timely manner.			
Task #	Task Description	Task ESD	Task ECD
3.1	Systems Requirement Review for Subsystem Parts	Feb 2027	Mar 2027
3.2	Vendor Identification and Selection Process	Mar 2027	Jun 2027
3.3	Procure Avionics, GNC, Telecom, and Thermal Systems Parts: (Cameras, Spectrometer, Sensors, Antennas, Radiators)	Jun 2027	Aug 2027
3.4	Procure Structure, Mechanism, Power Systems: (Nickel Titanium Wheels, Drill, RTG)	Aug 2027	Oct 2027
3.5	Inspection, testing and Documentation of Parts procured	Oct 2027	Nov 2027



Work Package #4 Comprehensive Performance Testing (CPT)

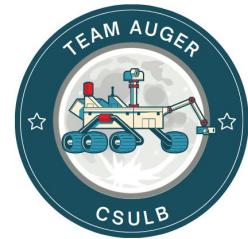


Deliverable	Completely assembled qualification unit, built with official hardware, along with verification of all requirements, and undergone comprehensive performance testing.	ESD ECD	Feb 2028 Jul 2029
Description: A qualification will be constructed using its intended hardware along with any changes made to the original design. It will undergo testing to failure, as well as verification testing, and any final changes will be made.			
Task #	Task Description	Task ESD	Task ECD
4.1	Assembly of the Qual Subsystems.	Feb 2028	Apr 2028
4.2	Testing of the Qual Subsystems.	Apr 2028	Jun 2028
4.3	Integrate all tested subsystem into a complete Qual unit.	Jun 2028	Sep 2028
4.4	Rover's structure stress testing.	Jan 2029	Feb 2029
4.5	Drill's performance and endurance testing.	Feb 2029	Apr 2029
4.6	Verify subsystem and overall requirements.	Apr 2029	May 2029
4.7	Final review of all acquired data from testing.	May 2029	Jun 2029
4.8	Final design adjustments (if necessary) on design based on Qual testing results.	Jun 2029	Jul 2029



Work Package #5

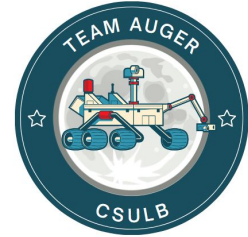
Limited Flight Unit Performance Test



Deliverable	Flight unit, identical to the qualification unit, undergone limited performance testing, and is prepared for launch.	ESD ECD	Aug 2029 Nov 2030
Description: A flight unit, identical to the qualification unit (along with any changes made) will be constructed as well as undergo limited testing to verify its performance. Preparations for launch will then be made.			
Task #	Task Description	Task ESD	Task ECD
5.1	Assembly of all subsystems for Flight Unit	Aug 2029	Dec 2029
5.2	Testing of all subsystems for Flight Unit	Jan 2030	Mar 2030
5.3	Integrate all subsystems into the Flight unit	Mar 2030	Jun 2030
5.4	Limited testing to verify the performance of the Flight unit	Jun 2030	Aug 2030
5.5	Final evaluation of test data acquired	Aug 2030	Oct 2030
5.6	Flight Close out	Oct 2030	Nov 2030



Planned Software



Flight Software

- C++ (Avionics, Telecommunication)
- Autonomous Operating System, AOS (GNC)



Ground Support Software

- Systems Tool Kit (Space Mission Planning & Simulation)
- MATLAB/Simulink (Data analysis & Visualization)
- RadCAD Thermal Desktop (Thermal Analysis Tool)
- LabView (Developing custom control interfaces)
- Trello (Project Management Tool)
- Siemens NX, SolidWorks (Mechanical Design, Tooling)
- GDB (C++ Debugger)



LabVIEW





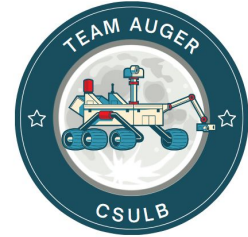
Where Will The Work Be Done



Design & Business	Subsystem Facilities	Testing	Assembly
Design: CSULB ECS Long Beach, CA	Structures: NASA Ames Research Center Mountain View, CA	Environmental Testing: NASA Goddard Space Flight Center Greenbelt, MD	System Integration: NASA Johnson Space Center Houston, TX
Design: CSULB Horn Center Long Beach, CA	Telecommunications: NASA Jet Propulsion Lab Pasadena, CA	Data Transmission Testing: NASA Jet Propulsion Lab Pasadena, CA	System Integration: Northrop Grumman San Diego, CA
Business: Remote Meetings held via ZOOM	Thermal: NASA Goddard Thermal Branch Greenbelt, MD	Neutron Spectrometer Accuracy Testing: NASA Johnson Space Center Houston, TX	System Integration: Michoud Assembly Facility New Orleans, LA
	Avionics / GNC: NASA Ames Research Center Mountain View, CA	Regolith Simulation Testing: Astrobotics Lunar Regolith Lab Pittsburgh, PA	System Integration: NASA Jet Propulsion Lab Pasadena, CA
	Mechanisms: NASA Johnson Space Center Houston, TX	Regolith Drill Testing NASA Johnson Space Center Houston, TX	
	Power: NASA Jet Propulsion Lab Pasadena, CA	Power Compatibility Testing: NASA Jet Propulsion Lab Pasadena, CA	



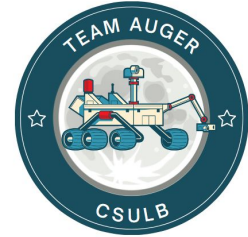
Deliverables



Work Package	Description	ECD
1	A completely built and tested flat-sat that is fully functional and capable of proper communication between each subsystem throughout the entire vehicle.	Nov 2025
2	A constructed engineering model, built with commercial parts, and has been fully tested, with collected data that has been analyzed to govern design changes if necessary.	Feb 2027
3	Necessary hardware from manufacturers procured and has been prepared for the assembly of the qualification and flight unit.	Dec 2027
4	A completely assembled qualification unit, built with official hardware, along with verification of all requirements, and has undergone comprehensive performance testing.	Jul 2029
5	Flight unit, identical to the qualification unit, undergone limited performance testing, and is prepared for launch.	Nov 2030



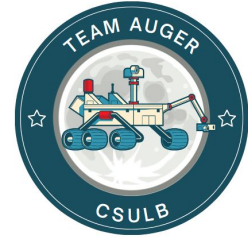
Milestones



Scheduled Task ESD	Scheduled Task ECD	Milestone
Jan 2025	Nov 2025	Work Package #1 - Flat-Sat integration test
Jan 2026	Nov 2026	Work Package #2 - Full up comprehensive performance test of the EM
Feb 2027	Dec 2027	Work Package #3 - Obtain all parts for Qual and Flight unit from manufacturers
Feb 2028	July 2029	Work Package #4 - Qual unit drilling mechanism performance test
Aug 2029	Nov 2030	Work Package #5 - Final evaluation of assembled Flight unit



Next Activities



Final Design

Finalize rover design including the structural components as well as integration of subsystem parts.

Testing

Testing of the functionality and feasibility of every subsystem component and the collaboration of them together.

Verifying

Evaluate and analyze testing results, verifying requirements, and ensuring optimal functionality.

Integration

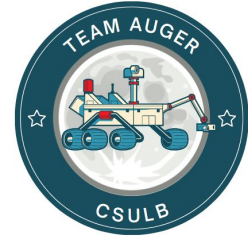
Make necessary changes according to the testing results and analysis.

CDR

Complete and final design along with calculations and analysis to support claims and decisions.



Issues and Concerns



Concerns :

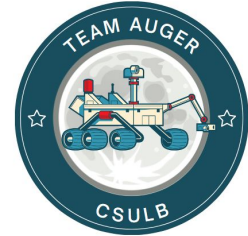
- Drilling mechanism causing sublimation
- Unpredictable soil behavior
- Being able to withstand the extreme cold temperatures

Issues :

- There are no current issues that prevent the completion of this project.



Team Leaders Assessments

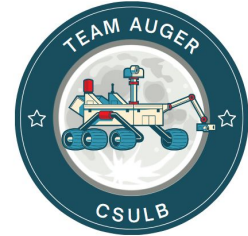


Currently, one of the biggest concerns is the potential sublimation that may occur during drilling. To mitigate this concern, we are committed to thoroughly test our drilling mechanism to ensure its reliability and minimize potential risks.

I have confidence in Team Auger's capabilities to design and develop our rover and meet our mission requirements. While we acknowledge the potential risks and challenges that may arise, we are committed to addressing these by continuously reviewing our design, conducting in-depth research, rigorous testing, and maintaining an open-minded approach. Through these efforts, we are assured success in creating a rover capable of effectively searching, drilling, and storing ice water at the South Pole of the Moon.



Q & A



Thank you, any questions or concerns?